MediConnect BD – Client Requirements

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We need a web platform where patients can find doctors, book appointments, consult by video call and receive digital prescriptions, built and live within 12 working days.

1. About Us

MediConnect BD is a healthcare startup in Dhaka. Patients today wait for hours in chambers or call serial numbers that never pick up, and people outside Dhaka struggle to reach specialists. We want a platform where patients book a confirmed time slot with a doctor and, when needed, consult online by video.

Who will use the platform

- **Visitors:** anyone browsing doctors without an account
- **Patients:** registered users who book and attend appointments
- **Doctors:** verified doctors who set their schedule and consult patients
- **Admin:** our team who verify doctors, manage specialties and monitor the platform

2. What Patients Need

- Search doctors by name, specialty (Medicine, Cardiology, Pediatrics, etc.), hospital, area and consultation fee
- See a doctor's profile: photo, degrees, experience, chamber address, fees, languages, rating and reviews
- See available time slots for the next 7 days and book one, choosing **in-person (chamber)** or **video consultation**
- Pay for video consultations online before the appointment; in-person visits can be paid at the chamber
- Get a booking confirmation by email, and a reminder before the appointment
- Join the video call from the browser at the appointment time, with no app to install, and chat or share files (old reports, photos) during the call
- Upload medical reports to their profile so the doctor can see them before the visit
- Receive a digital prescription after the consultation and download it as PDF
- See upcoming and past appointments, cancel or reschedule an upcoming one

- Rate and review a doctor after a completed appointment
- Manage family members (book for a parent or child from one account)

3. What Doctors Need

- Register with BMDC registration number and certificates; they can work only after admin approval
- Set a weekly schedule (for example, Sunday–Thursday 5 PM–9 PM), slot length (10, 15 or 20 minutes) and separate fees for chamber and video
- Block specific dates for leave or emergencies; patients already booked on those dates must be notified
- See today's appointment queue and upcoming appointments
- See the patient's past visits, uploaded reports and previous prescriptions before the consultation
- Start the video call from the dashboard and see when the patient has joined
- Write a prescription during or after the visit: diagnosis, medicines (name, dose, timing, duration), tests and advice, with medicine name suggestions while typing
- Mark an appointment as completed or no-show
- See their earnings by day and month

4. What Our Admin Needs

- Review doctor applications, see uploaded documents, and approve or reject with a reason
- Suspend a doctor or patient account
- Manage the list of specialties and hospitals
- See all appointments and payments, filter by date, doctor and status
- Handle refunds for cancelled video consultations
- Remove abusive reviews
- A dashboard: total appointments, today's appointments, revenue, new patients, most booked specialties and top doctors

5. Business Rules

- One slot can be booked by only one patient, even if several patients try to book the same slot at the same moment.
- When a patient starts booking a video slot, hold it for 10 minutes. If payment is not completed, the slot becomes free again automatically.
- Patients can book up to 7 days ahead, and not less than 1 hour before the slot.
- A patient cannot have two appointments at overlapping times.
- Cancellation: more than 24 hours before → full refund; 2–24 hours before → 50% refund; less than 2 hours → no refund. If the doctor cancels, the patient always gets a full refund.
- Reminders go out 24 hours and 30 minutes before the appointment.
- The video room opens 5 minutes before the slot and only the booked patient and that doctor can join it.
- A prescription can be edited for 24 hours after the visit; after that it is locked. Every prescription has a unique ID that can be verified.
- Only the patient and their doctors can see the patient's reports and prescriptions. Admin cannot read medical content.
- Patients can review a doctor only after a completed appointment, once per appointment.
- The platform keeps a 10% commission on each paid consultation.
- All times are shown in Bangladesh time.

6. Technical Preferences

- Frontend: React with TypeScript
- Backend: Node.js with Express and TypeScript
- Database: MySQL
- Video calls must run in the browser (WebRTC or a WebRTC-based service)
- Online payment through SSLCommerz (sandbox is fine for now)
- Reminders, emails and slot release must happen automatically in the background, without slowing down the website
- The whole project should start with one command on a new machine (for example, Docker)
- Must work well on mobile phones; many patients will join video calls from a phone

- Medical data must be kept secure and private
- Live on the internet at the end

7. Not Needed Right Now

Native mobile app, SMS (email is enough), lab test booking, medicine home delivery, insurance claims, bKash or Nagad payment, and Bangla language support. We may ask for these later.

8. Timeline

12 working days. We would like to see progress at these points:

Day	What we want to see
3	Your requirement analysis, database design and planned APIs; project runs with one command; login for all three user types
6	Doctors can set schedules; patients can search doctors and see free slots
9	Booking with slot hold, online payment, email confirmation and automatic reminders
12	Video consultation, prescription PDF, reviews, admin dashboard, all live on the internet

9. What We Expect From You

- Your own analysis of these requirements: questions, assumptions and user stories
- Database design (ERD) with your reasoning
- API documentation
- An explanation of how you prevent two patients booking the same slot, with a test that proves it
- Source code on GitHub with regular commits, and automated tests for booking and cancellation
- Live link with demo logins for one patient, one doctor and one admin

- A README explaining how to run the project
- A short demo, including a live video call between a patient and a doctor

If anything here is unclear, ask us before building. Where we cannot answer quickly, write down your assumption and continue.